REPORT

Visual Content-Based Music Recommendation

Miracle Aligners

Introduction to AI & DS - Group Project Oral Session

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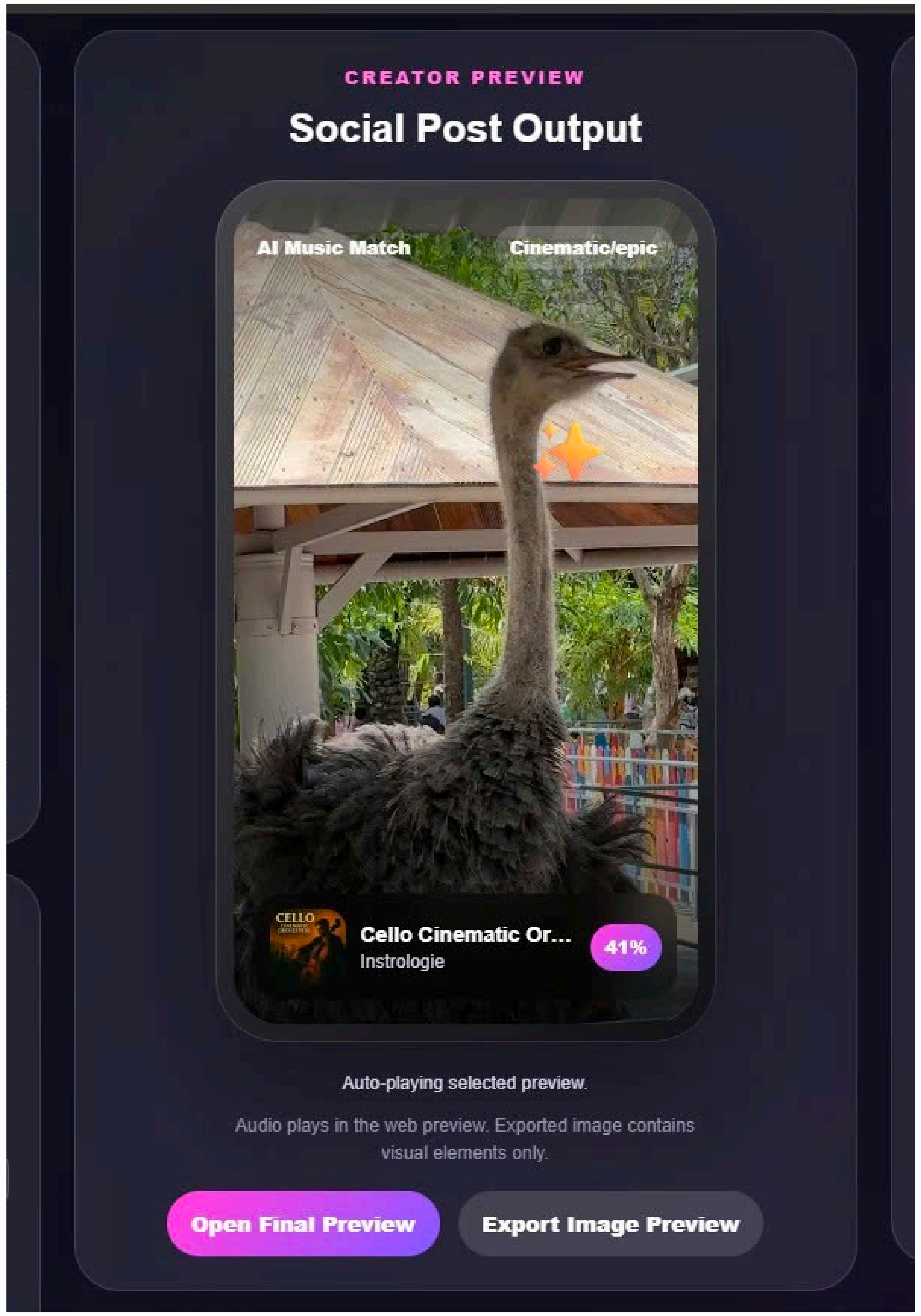
INTRODUCTION

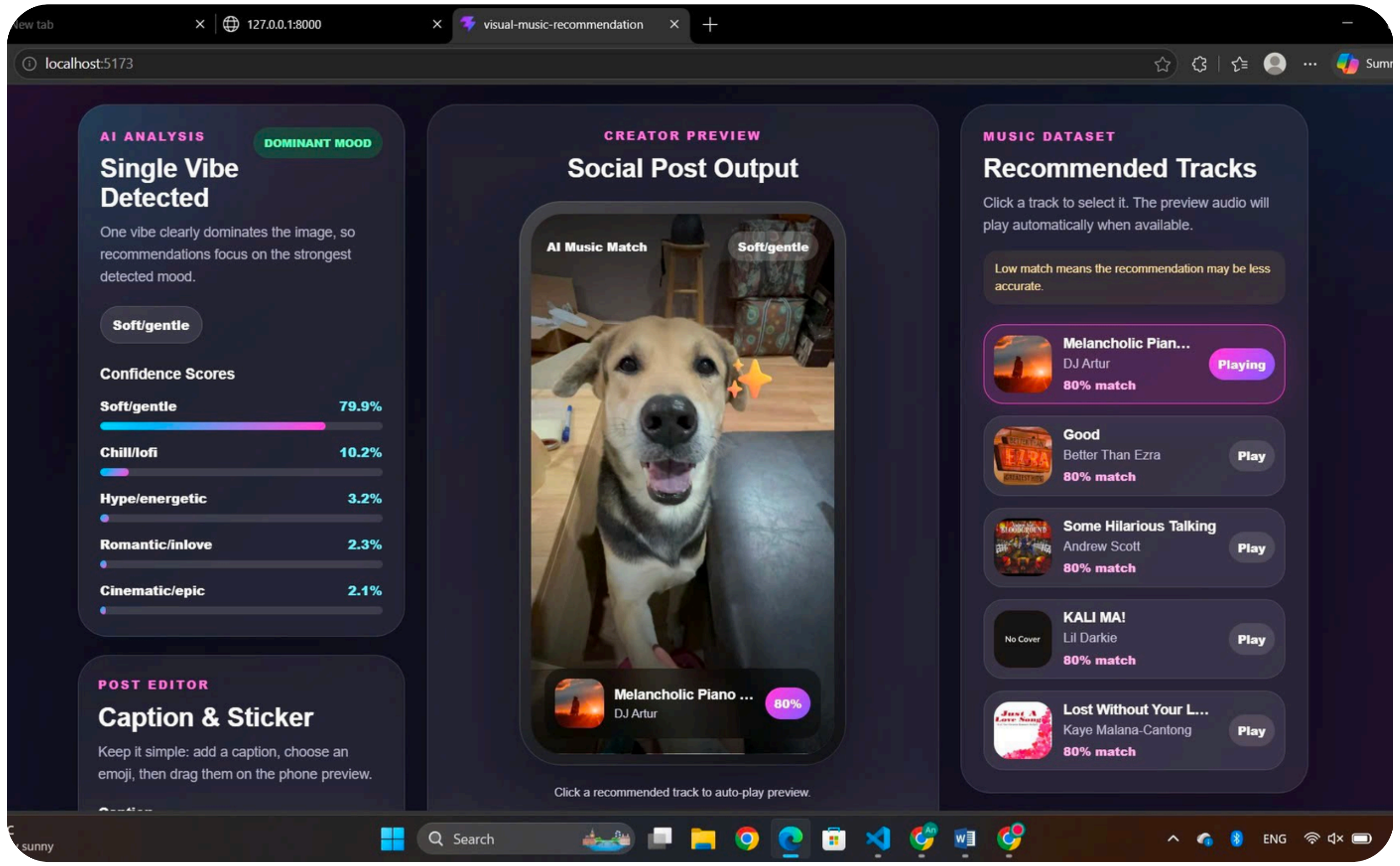
Visual Content-Based Music Recommendation System

Research Question:

Can an image alone recommend music?



PROBLEM STATEMENT



Challenges

- Depend on user interaction history
- Cold-start problem
- Large datasets required

Our Solution

Content-Based Music Recommendation

PROPOSED SOLUTION

Instead of predicting music from user history, we introduce an intermediate semantic layer called **Vibe**.

Pipeline:

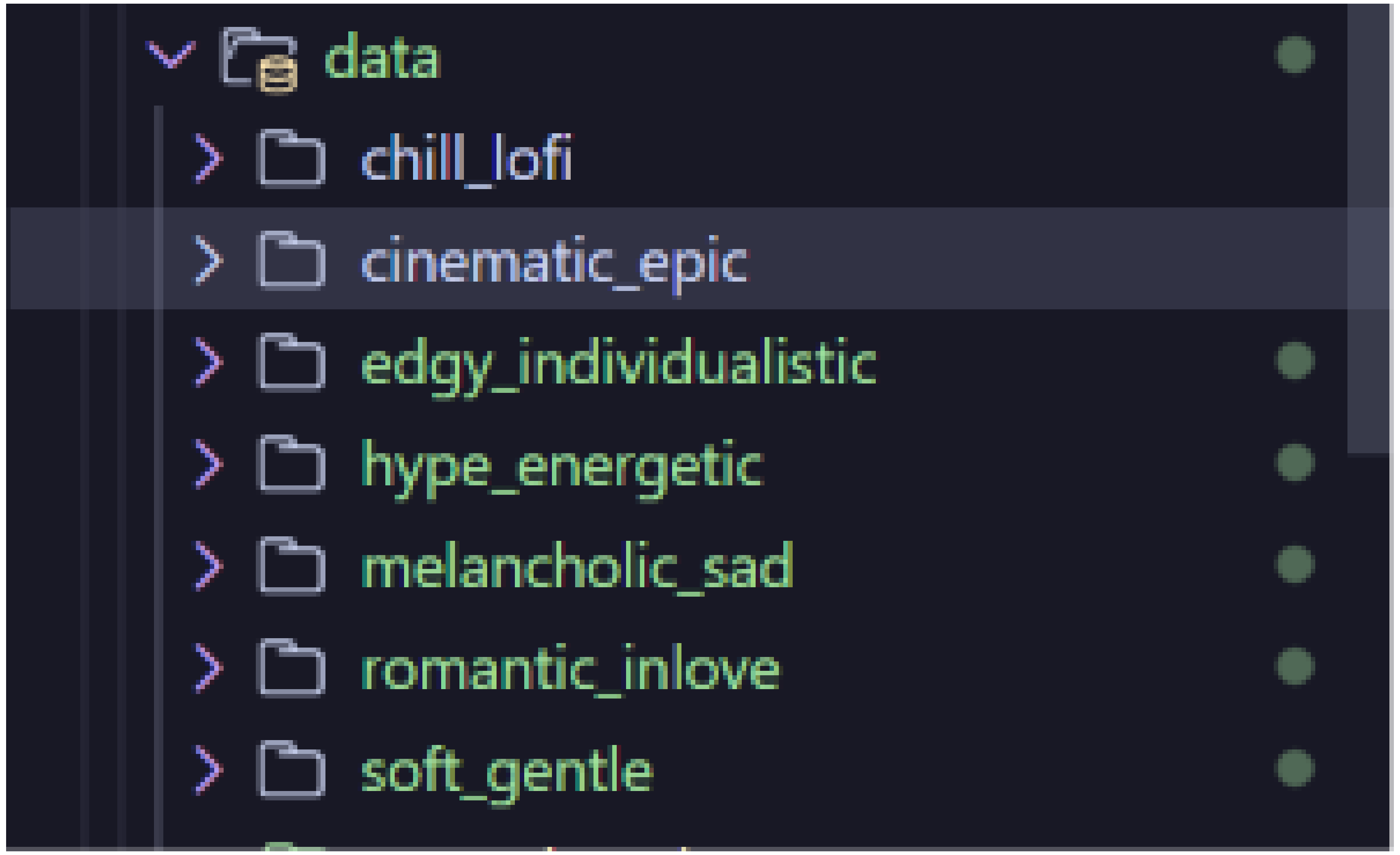
Image

Vibe

Music

Vibe bridges visual and musical features.

Dataset



~1300 images

Category	Playlist/song URLs
chill / lofi	Chill Tracks | Spotify Playlist
chill / lofi	Chill lofi day - Chill beats for Focus, Study & Relaxation - playlist by H-MUSIC FR | Spotify
chill / lofi	beats to relax/study
hype / energetic	Hype Workout Mix | Spotify Playlist
hype / energetic	Hype Workout Bangers | Pre Game Hype Music - PumpUp Music - playlist by music moods | Spotify
edgy / individualistic	Modern Darkwave, Post-Punk, Synth & EBM | RELEASES of 2019 - playlist by ZAUBER | Spotify
chill / lofi	chill lofi study beats | Spotify Playlist
hype / energetic	Hype High Energy Mix | Spotify Playlist
hype / energetic	HYPE EDM (Club, Festival, and Gym Hits) - playlist by Luke Gullickson | Spotify
hype / energetic	Hype Hits: Best of Hip Hop - playlist by ESPN Music | Spotify
melancholic / sad	Sad indie - playlist by Mattias | Spotify
melancholic / sad	ACOUSTIC/FOLK MELANCHOLY - playlist by Fashionpolice | Spotify
melancholic / sad	neofolk/dark folk/acoustic/melancholy... - playlist by hohenangstt | Spotify
melancholic / sad	Acoustic Pop-Punk Melancholy by Pretending Serenity
romantic / in love	https://open.spotify.com/playlist/3zRajLtxlpOSgMwdy8dsef
romantic / in love	Relaxing Music 🎵 New Age Piano Music for Sleep/Study/Relax - playlist by The Soul of Wind | Spotify
romantic / in love	Nhac tình yêu😊 💔 - playlist by Xu Yang Hồ | Spotify

~200 - 300 songs per vibe

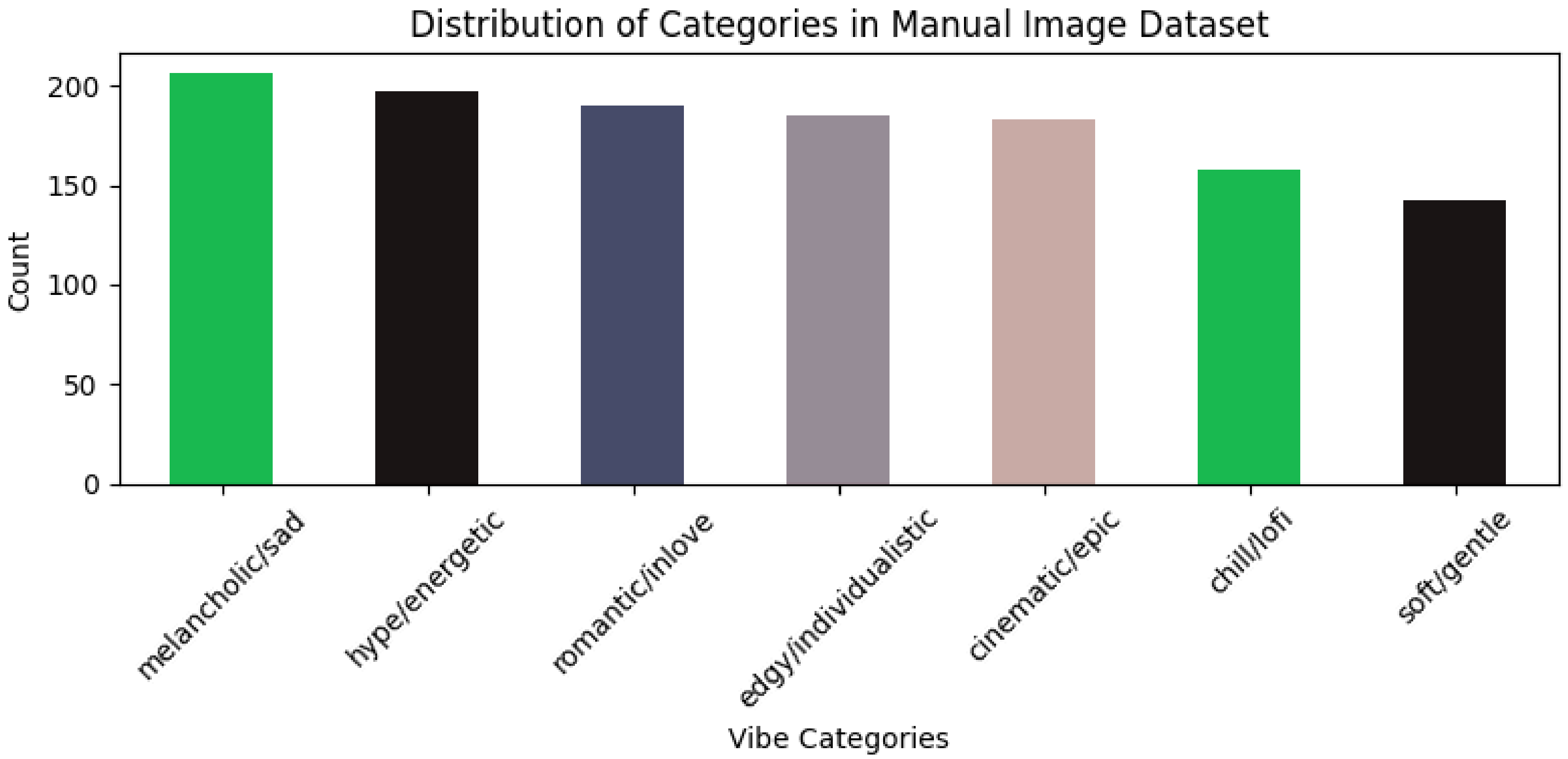
Dataset

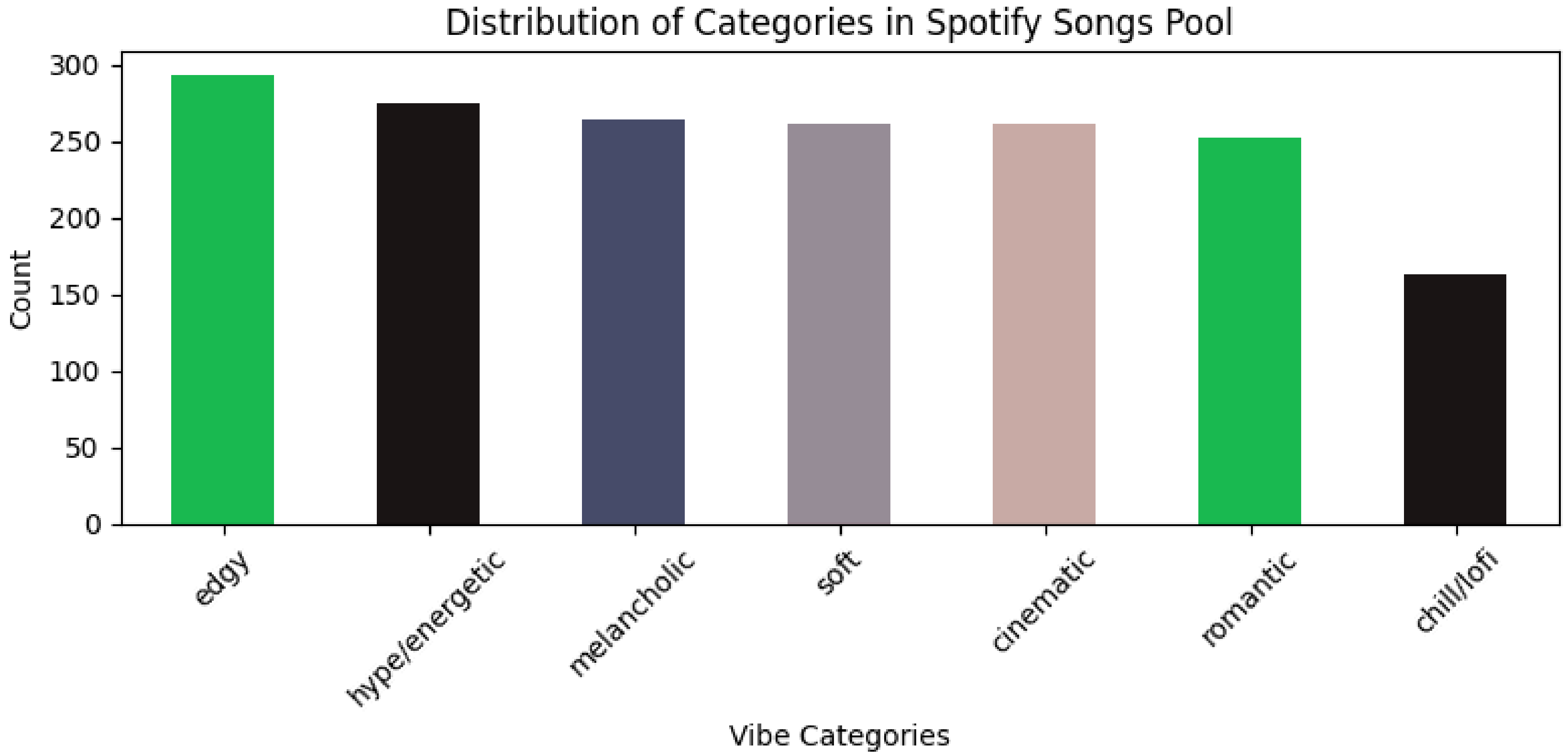
```
=IF(COUNTA(B2:D2)<3, "Waiting for all votes...", IF(C2=D2, C2, IF(C2=E2, C2, IF(E2=D2, E2, "DISCARD_TIE"))))
```

	A	B	C	D	E	F
1	Folder	Number	Vy	Trâm	Linh	Final
2	chill/lofi	1	chill/lofi	chill/lofi	chill/lofi	chill/lofi
3	chill/lofi	2	chill/lofi	chill/lofi	chill/lofi	chill/lofi
4	chill/lofi	3	chill/lofi	cinem...	chill/lofi	chill/lofi
5	chill/lofi	4	chill/lofi	chill/lofi	chill/lofi	chill/lofi
6	chill/lofi	5	chill/lofi	chill/lofi	chill/lofi	chill/lofi
7	chill/lofi	6	chill/lofi	chill/lofi	chill/lofi	chill/lofi
8	chill/lofi	7	chill/lofi	chill/lofi	chill/lofi	chill/lofi
9	chill/lofi	8	chill/lofi	cinem...	chill/lofi	chill/lofi
10	chill/lofi	9	chill/lofi	chill/lofi	chill/lofi	chill/lofi
11	chill/lofi	10	chill/lofi	chill/lofi	chill/lofi	chill/lofi
12	chill/lofi	11	chill/lofi	chill/lofi	chill/lofi	chill/lofi
13	chill/lofi	12	chill/lofi	chill/lofi	chill/lofi	chill/lofi
14	chill/lofi	13	soft/ge...	chill/lofi	chill/lofi	chill/lofi
15	chill/lofi	14	chill/lofi	chill/lofi	chill/lofi	chill/lofi
16	chill/lofi	15	chill/lofi	chill/lofi	chill/lofi	chill/lofi
17	chill/lofi	16	chill/lofi	chill/lofi	chill/lofi	chill/lofi
18	chill/lofi	17	chill/lofi	chill/lofi	chill/lofi	chill/lofi
19	chill/lofi	18	chill/lofi	chill/lofi	chill/lofi	chill/lofi
20	chill/lofi	19	chill/lofi	chill/lofi	chill/lofi	chill/lofi
21	chill/lofi	20	chill/lofi	chill/lofi	chill/lofi	chill/lofi
22	chill/lofi	21	chill/lofi	chill/lofi	chill/lofi	chill/lofi
23	chill/lofi	22	chill/lofi	chill/lofi	chill/lofi	chill/lofi
24	chill/lofi	23	chill/lofi	chill/lofi	chill/lofi	chill/lofi
25	chill/lofi	24	chill/lofi	chill/lofi	chill/lofi	chill/lofi

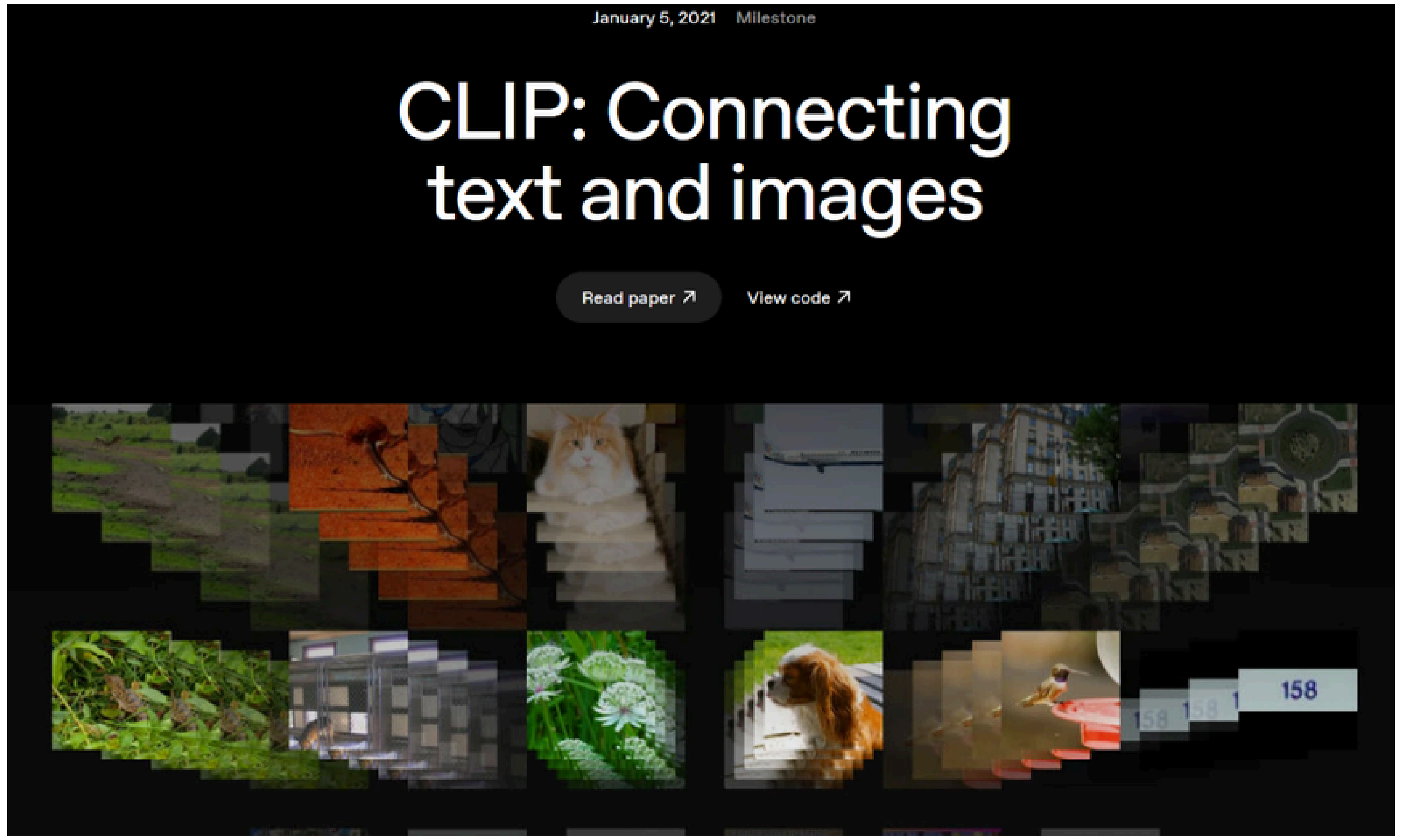
songs url | images evaluation

Dataset


Distribution of Categories in Manual Image Dataset
Count
0
50
100
150
200
melancholic/sad
hype/energetic
romantic/inlove
edgy/individualistic
cinematic/epic
chill/lofi
soft/gentle
Vibe Categories


Distribution of Categories in Spotify Songs Pool
Count
0
50
100
150
200
250
300
edgy
hype/energetic
melancholic
soft
cinematic
romantic
chill/lofi
Vibe Categories

Training



Why Zero-Short?

→ It understands abstract vibes better than traditional CNNs

Logic & Adapter #1

The Softmax Trap

51% + 51% = 102% ?

→ Absolute threshold breaks multi-vibe detection

Logic & Adapter #2

Margin-Gap Optimization

$$\text{Gap} = |\, \text{Score}_1 - \text{Score}_2 \,|$$

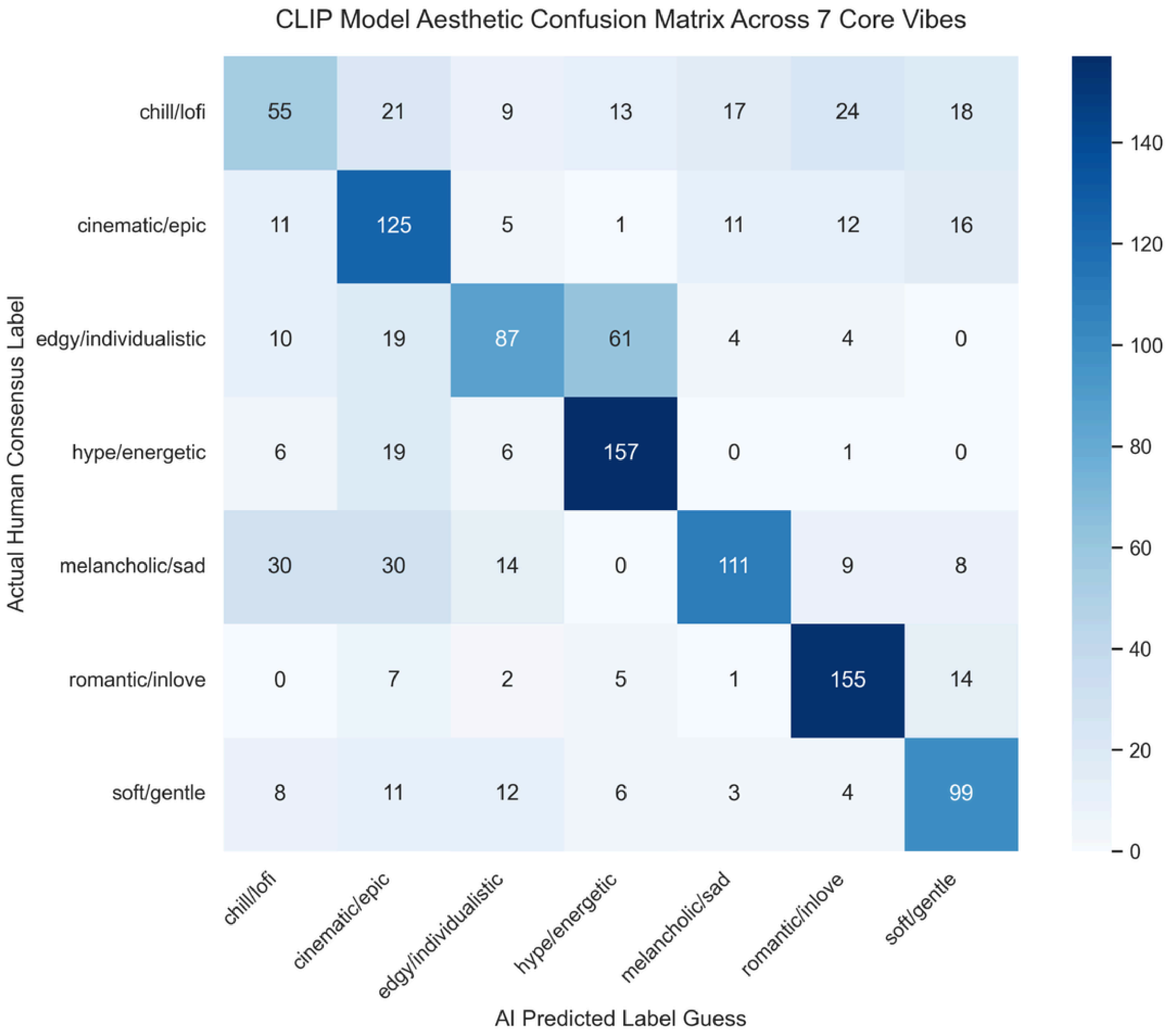
→ Our logic detects when a photo is **hybrid** of 2 moods

Performance #1

```
❌ Folder: soft/gentle | Img: 174 -> AI guessed 'chill/lofi', Human column said 'soft/gentle'
✅ Folder: soft/gentle | Img: 175 -> AI Matched Human (soft/gentle)
❌ Folder: soft/gentle | Img: 176 -> AI guessed 'cinematic/epic', Human column said 'soft/gentle'
❌ Folder: soft/gentle | Img: 177 -> AI guessed 'edgy/individualistic', Human column said 'soft/gentle'
❌ Folder: soft/gentle | Img: 178 -> AI guessed 'cinematic/epic', Human column said 'soft/gentle'
✅ Folder: soft/gentle | Img: 179 -> AI Matched Human (soft/gentle)
✅ Folder: soft/gentle | Img: 180 -> AI Matched Human (soft/gentle)

--------------------------------------------------------
📊 LIVE ACCURACY TESTING REPORT COMPLETE!
➖ Total Discarded Ties (Skipped): 0
🎯 Clean Images Evaluated: 1241
✨ Correct AI Classifications: 789
🏆 FINAL PRESENTATION GRADE ACCURACY: 63.58%
(venv)
@igdngl /d/00 VGU/YEAR 2/Introduction to Data Science and AI/group-project/visual-music-recommendation/backend
feature/backend-ai)
>
feature/backend-ai*   ⊗ 0 ⚠ 0   -- INSERT --|154 fewer lines; before #2  41 seconds ago     Ln 8, Col 48   Spaces: 4   UTF-8   CRLF
```

Performance #2



THANK YOU

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